

Develop a Design for a Digital Outcome

Task 1: Purpose and End-Users Requirements

What context did you come up with?

Students in New Zealand often struggle to retain knowledge over the extended period between first learning and the November NCEA external examinations. While many invest significant time in revision, this effort is often unstructured and relies on passive strategies such as rereading or summarising, which research shows are relatively ineffective for long-term retention. As a result, students who initially understand content, particularly topics taught earlier in the year, often fail to recall it accurately by the time of assessment. This leads to outcomes that do not reflect their true understanding or capability. This issue is significant, as national data shows that a substantial proportion of students do not achieve NCEA Level 2 and Level 3. The problem is not simply effort, but the absence of systems that support consistent, long-term memory retention. Therefore, this project aims to develop a digital outcome that improves how students manage and maintain their knowledge over time, supporting more effective preparation for end-of-year external examinations.

Who are the end-users of your website?

The primary end-users are NCEA secondary school students in New Zealand, particularly those in Years 12–13 preparing for external examinations. These students typically manage multiple subjects simultaneously and often lack structured systems to organise long-term revision across the academic year. The tool is especially targeted at students who are willing to study but struggle with consistency, retention, and knowing what to revise at any given time.

What is the purpose of the outcome?

The purpose of the outcome is to support long term retention of academic content, guide students on what to review and when, using a structured spacing system. The tool focuses on scheduling revision in a way that promotes consistent engagement and reduces forgetting over time, which can improve student preparedness for external exams.

What are the end-user requirements?

End-users (students from year 12-13) require a system that is simple, fast to use, and does not add additional workload to their study routine.

1. **Requirement 1 is that the tool shows students what to study without them having to decide.**
2. **Requirement 2 is that the tool be usable in under 30 seconds for a single review.**
3. **Requirement 3 is that the interface is clean and uncluttered.**
4. **Requirement 4 is that the tool track progress visibly.**
5. **Requirement 5 is that the tool work across different devices.**

Task 2: Research

Website PMI

Links	Plus	Minus	Interesting
Quizlet https://quizlet.com/nz/	- Card-based grid layout that have clean, scannable structure - - Strong 2-colour palette (blue/white)= visual consistency - Single-focus "Learn" mode removes all other competing elements during review	- Six study mode options shown same time, hard decision dilemma before studying begins - Paywall padlocks scattered throughout that can erode trust for free users - No visible indicator of <i>when</i> content was last reviewed or is next due	- Progress shown within a session but disappears entirely between sessions - Colour feedback (green/red) during review creates instant emotional signal which can worth adapting for status coding
Anki https://ankiweb.net/	- Web login is provided to essentials like two inputs, one button, zero distraction - No advertising, no upsells, no promotional clutter anywhere - Review interaction itself is	- Landing page shows nothing about what the tool does, for example a first-time visitor has no reason to continue - Requires learning a technical vocabulary (decks,	- Desktop app uses a single "cards due today" number as the entire dashboard -> the most compressed possible expression of daily priority - Colour-coded response buttons (red = Again, green = Easy) communicate urgency with no clear labels

	elegantly minimal: one question, one button, four responses	note types, ease factors) before first review begins - Statistics and progress graphs buried three menus deep, the students who need motivation most never see their own improvement -Super messy and confusing design	for new users
Study Time https://studytime.co.nz/	- NCEA Level 1/2/3 resources is immediately familiar to NZ students, zero orientation cost -Straightforward/simple white and blue palette feels local and student-friendly without being corporate - "Start Here" pathway bypasses complex navigation for new users	- Homepage overloaded is quite hero banner, three/four nav bars, articles, videos, and store promotions compete simultaneously - Every student sees an identical page regardless of year level or prior visits - No interactive features or return-visit mechanism,	- Resource cards use consistent thumbnail + title format throughout predictable structure reduces cognitive effort - Most-referenced NZ study site contains zero retention scheduling,that can directly confirming the structural gap my tool fills

		students come back only when they remember to	
Duolingo https://www.duolingo.com/	- Single hero CTA on a clean homepage, the page does one thing with max clarity - Completion ring fills as daily goal is met, communicates "nearly done" without any text - Positive empty-state design ("You're all caught up!") turns zero-task screens into a reward moment	- Gamification elements accumulate over time, that can make the main screen progressively more cluttered - Notification strategy is aggressive by default (send too much notifications), disturbed - Hearts/lives system penalises errors, actively harming low self-efficacy students	- One default daily goal that most users never change, nudge design shapes behaviour through defaults rather than instruction - Web version is simpler than the app, -> the risk of designing app-first and treating web as secondary
Education Perfect https://www.educationperfect.com/	- Persistent colour-coded subject icons allow instant subject-switching without reading labels - Progress bars	- All tasks are teacher-assigned, students cannot self-initiate review - Leaderboard and accuracy data is class-visible, can conflicting	- Student engagement publicly tracked and reported to teachers , a case study in exactly what not to do for a tool targeting students who struggle academically

	within lessons use familiar completion percentage format students already understand - Responsive layout functions correctly across mobile and tablet	with my privacy and self-efficacy requirements - Accuracy percentage (right/wrong in one sitting) is presented as progress, _> short-term performance with long-term retention	- AI feedback on responses is an emerging convention, confirming automated personalised feedback is the direction educational tools are heading
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Task 3: Investigate conventions

Convention 1: (what is the convention?) - Persistent color coded status indicators

Every researched site uses color to communicate status without requiring the user to read text. Anki uses red/orange/green on response buttons. Quizlet uses green/red for incorrect/correct feedback. Education perfectly uses color-tagged subject icons. Duolingo uses green fills on completion rings. The convention is consistent across most platforms where green signals completion or safety, amber/orange signals action required, red signals urgency or error. This color language is so established across many educational tools that users arrive already knowing what it means with no learning cost. Link to match between system and real world.

Convention 2: (what is the convention?) Single primary nav structure (top bar or left sidebar)

Most websites place their primary nav in one of two positions: a fixed horizontal bar at the top of every page (quizlet, studytime, ep, duoling) or a fixed left side panel (anki desktop). This convention exists because users instinctively scan nav in these positions, placing it anywhere forces reorientation on every page. A top bar is more common in mobilefirst tools, the left sidebar appears in tools with many subject categories. it can keep nav accessible without occupying much of the central content area. -> Consistency and standard

Convention 3: (what is the convention?) Progress indicator tied to a daily/session goal

Every education site provides some form of visual progress tied to immediate, completable goals rather than a long term outcome. Like duolingo uses a filling ring for daily XP goal. Anki shows a “cards remaining today” countdown, Education perfect shows percentage completion within a lesson. A daily goal with a visible progress indicator can give users some purposes and sort of rewards that make their experiences more entertaining. A long term retention outcome are too delayed, took too much time, to sustain daily engagement on their own

Convention 4: (what is the convention?) Card based layout for content units

Individual pieces of content, like study sets (quizlet), topics(anki), resources(studytime), lessons (duolingo) or tasks(ep) are displayed as self contained rectangular cards rather than lists or tables. This is like a universal convention in educational web tools as cards create visual boundaries between items so they can make dense info scannable without reading every word. Cards also scale predictably based on the devices, they stack vertically on mobile and spread into grids on desktop without breaking the layout. -> aesthetic and minimalist

Which of the conventions will I incorporate into my website?

I will use most of the common interface conventions I identified during research into existing educational platforms. I will use a card-based layout to display each topic clearly, because NCEA students managing five or six subjects at the same time need to be able to scan information quickly without having to read every word on the page.

I will also use colour-coded status indicators to help students quickly recognise which topics need review, because research into existing tools like Quizlet and Anki showed that colour is already used this way across educational platforms and students arrive already knowing what red, amber, and green means without needing to read anything. This is important for my tool specifically because students need to be able to check their priorities in a short amount of time, so colour communication speeds up that process.

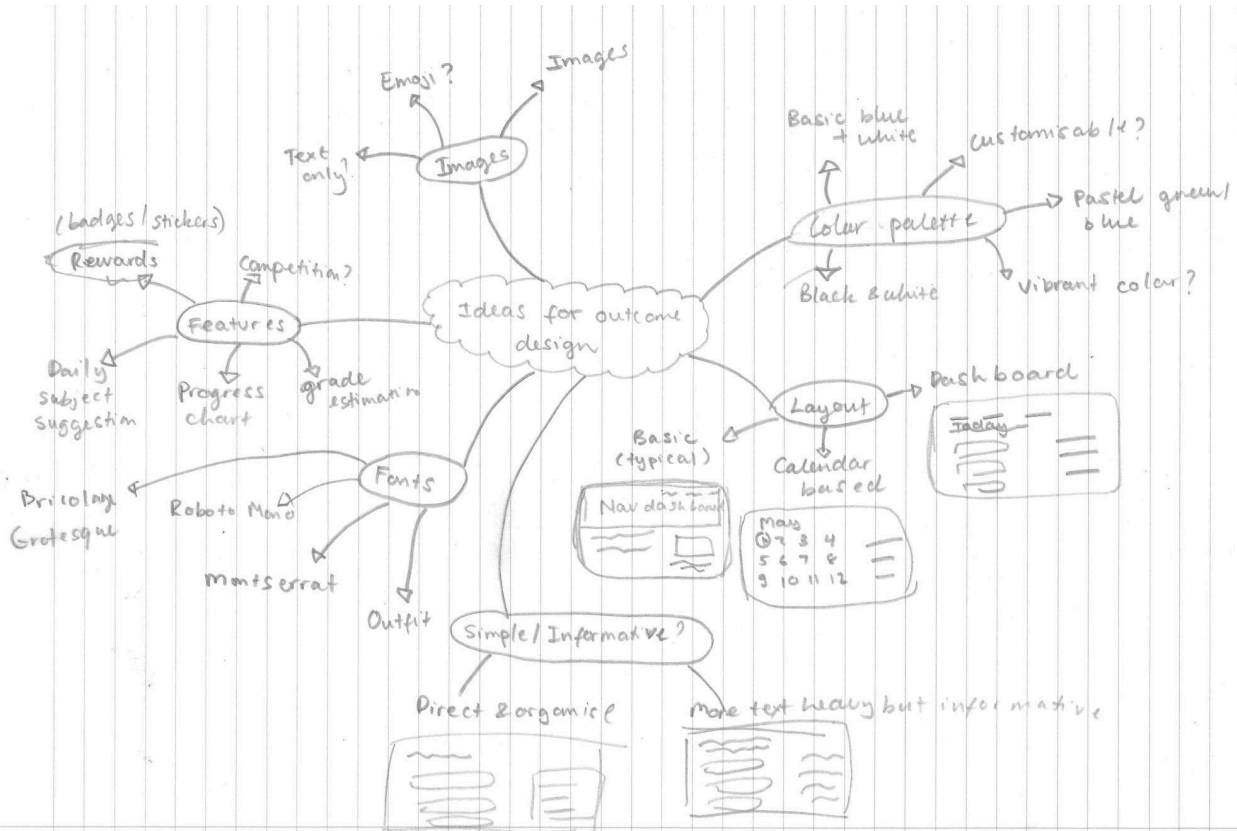
A simple navigation bar will organise the main sections of the tool, because low fidelity research into sites like Quizlet, StudyTime and Duolingo showed this is the standard placement for tools used on mobile and desktop.

The dashboard will include a visible progress indicator so students can track their revision over time, because a daily goal with a visible progress number gives users a sense of purpose and

makes the experience feel more motivating. Without this, students would have no way of knowing whether their effort over time is making a difference.

Task 4: Generate ideas

Include a screenshot of a mindmap here

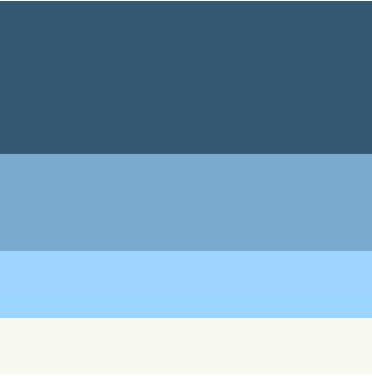
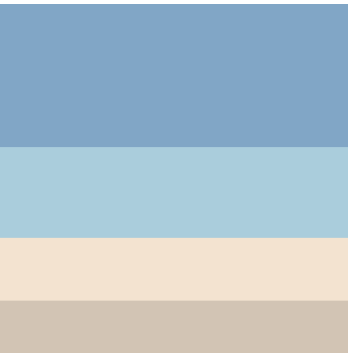


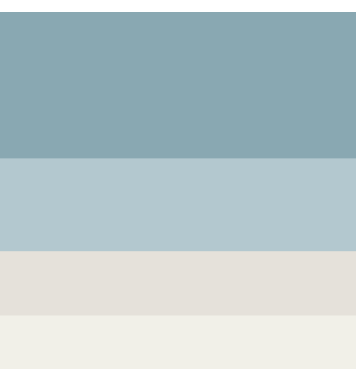
Design idea: Color palettes

Options (e.g. different colour pallets)

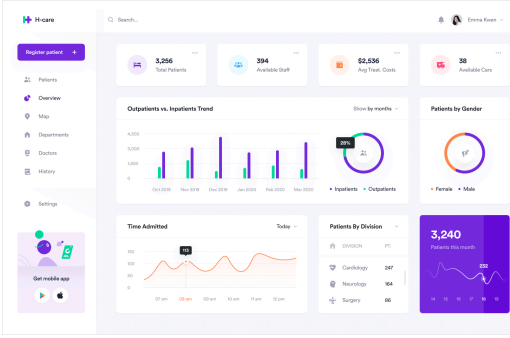
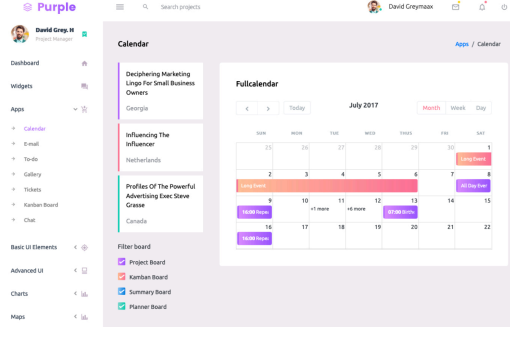
Votes (may choose more than 1)

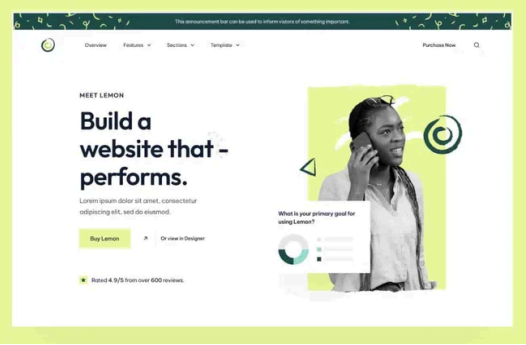
comments

	11	Stakeholder 2: The color is a bit too dark and depressing. The color makes me feel sad and will demotivate me to study. Stakeholder 4: I find it a good combo of all the blue. The color blue gives focus and calm which is great for studying the web. Stakeholder 7: provided the strongest readability and visual clarity, especially for quickly identifying priority tasks. The contrast of colours made status indicators more obvious and accessible
	1111	Stakeholder 3: I find it the perfect balance of blue (not too dark or light) Stakeholder 4: I like this one as well, but the last light brown color is not matching (the palette is not cohesive). Stakeholder 6: This felt calmer and more suitable for a study website that students would use regularly during stressful exam periods. The softer colours made the dashboard feel less overwhelming while still keeping important information clear. Stakeholder 8: best emotional balance and maintaining strong readability . it supports a calm and focused user experience without visually overwhelming the user. Stakeholder 5: felt less stressful and easier to look at for long periods of time. Since the website already reminds students about overdue topics, softer colours made the experience feel more encouraging

	111	Stakeholder 1: Monochromatic, calming experience for students Stakeholder 2: Favourite color, so i choose it Stakeholder 3: It's a nice shade, color are light and simple Stakeholder 8: Option 3 lacked strong contrast in some areas
	1	Stakeholder 3: the contrast is nice between different color (maybe too much if unable to balance) Stakeholder 4: red is too bright for study. It's giving me anxiety and pressure as I look at it. Stakeholder 6: looked too playful and less academic.
		Stakeholder 4: I don't like color. I don't think green and orange are connected. The color green out of content (irrelevant). I like the beige and blue combination though. Stakeholder 7: appeared less professional for an educational outcome.
	111	Stakeholder 1: Monochromatic, calming experience for students Stakeholder 3: The color is too light, could be boring Stakeholder 4: Color matching and calm. This can create soothing experience for student Stakeholder 6: looked dull and unmotivating

Design idea: Web layout

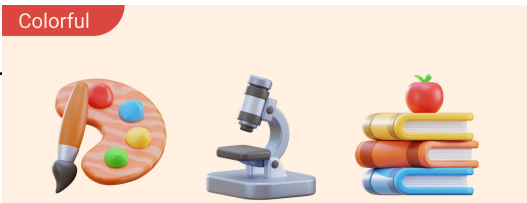
Options (e.g. different colour pallets)	Votes (may choose more than 1)	comments
 The screenshot shows a healthcare dashboard with a sidebar menu on the left containing 'Patients', 'Overview', 'Map', 'Departments', 'Doctors', 'History', and 'Settings'. The main area features several widgets: 'Total Patients' (3,256), 'Available Staff' (394), 'Avg Treat Costs' (\$2,536), and 'Available Cms' (38). There are three charts: 'Outpatients vs. Inpatients Trend' (a bar chart showing trends from Oct 2018 to Mar 2020), 'Patients by Gender' (a donut chart showing 21% for Females and 79% for Males), and 'Time Admitted' (a line chart showing trends from 10 am to 10 pm). Other widgets include 'Patients by Division' (listing Cardiology, Neurology, and Surgery) and a '3,240 Patients this month' badge.	111111	Stakeholder 2: because like include difference aspect, and stats motivate ne Stakeholder 3: I think this is the most useful by showing all the statistics Stakeholder 4: I find this is nice as well, as it can tell you your progress. It can be more motivating to see current stats and try harder to gain more % achieve etc. Stakeholder 7: everything important is visible immediately, including progress, overdue topics, and revision priorities. It feels motivating and efficient Stakeholder 5: Students can immediately understand what requires attention without excessive navigation. I like seeing all my progress and tasks immediately because it makes studying feel more manageable and motivating.
 The screenshot shows a project management dashboard with a sidebar menu on the left containing 'Dashboard', 'Widgets', 'Apps', 'Calendar', 'Email', 'To-do', 'Gallery', 'Notes', 'Kanban Board', 'Chat', 'Basic UI Elements', 'Advanced UI', 'Charts', and 'Maps'. The main area features a 'Calendar' widget with a 'Fullcalendar' view for July 2017. It includes a 'Taskboard' with a list of tasks like 'Deciphering Marketing Links For Small Business Owners', 'Influencing The Influencer', and 'Profiles Of The Powerful Advertising Exec Slave Grass'. There is also a 'Filter board' with options like 'Project Board', 'Kanban Board', 'Summary Board', and 'Planner Board'.	1	Stakeholder 2: We have a calendar similar to this layout (e.g. gg calendar, calendar on phone). It has no differentiation. Stakeholder 4: It could be great for time tracking like tracking your time what to do that day (not just today but beyond it as well so you know what to prepare. You can know directly when something is due Stakeholder 8: it does not strongly

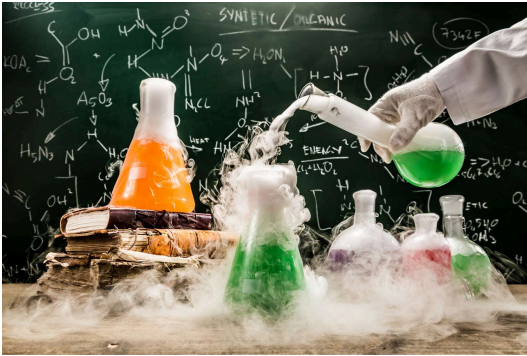
		differentiate between important and less important revision tasks. Stakeholder 6: could still be useful because students can directly see when reviews are scheduled and prepare ahead of time. However, it feels more like time management rather than an actual revision tool.
		Stakeholder 2 & 3: This layout is too simple. It can be really hard to organise and see your stats. And it make the site very long and lengthy Stakeholder 4: It looks like a normal web. It seems hard to navigate and having to zoom in/out stats which is distracting and inefficient. Stakeholder 7: layout feels too simple and difficult to organise once many subjects and topics are added. Important information becomes harder to prioritise visually. Stakeholder 5: looks like a normal website and may require too much scrolling and navigation, which feels inefficient for quick study sessions between classes.

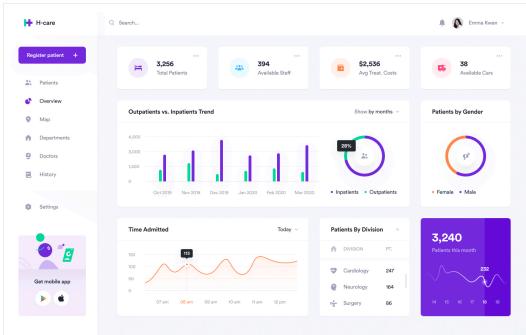
Design idea: Fonts		
Options (e.g. different colour pallets)	Votes (<i>may choose more than 1</i>)	comments
Regular 400 Daily priority list:	11	Stakeholder 3: It's a bit separate, a bit hard to read. and need to straight away. I think study fonts should be as simple as possible

		Stakeholder 7: the fonts looked too rigid Stakeholder 8: it felt too serious Stakeholder 5: I find it separated clearly and easy to read during long periods of screen Stakeholder 6: looks clean, simple, and readable for a study website
Regular 400 Daily priority list:	1111	Stakeholder 2: it's separate, easier to read Stakeholder 4: it's normal text, easy to read, looks right and clean Stakeholder 8: best readability for long-term screen use Stakeholder 7: looked too generic Stakeholder 5: font looks simple and visually appealing
Regular 400 Daily priority list:	11111	Stakeholder 4: It looks simple and cute Stakeholder 3: Too bold for normal text. I think it's messy, my eyes will hurt by reading it. Stakeholder 6: it's most digitally modern and remaining highly readable on both desktop and mobile devices Stakeholder 5: it looked the cleanest and easiest to read on a phone screen
Daily priority list:	11	Stakeholder 4: it's bold and easy to read Stakeholder 5: less readability due to its decorative appearance Stakeholder 7: look slightly childish

		for senior secondary students, and less the professional tone of the website. Stakeholder 8: it looked distracting and less professional to me
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Design idea: Images/icon		
Options (e.g. different colour pallets)	Votes (<i>may choose more than 1</i>)	comments
Black and white icons  shutterstock.com • 777148645	111	Stakeholder 3: These would work, might work depending on the web color. Stakeholder 2: may be dark and dull though
 Color icons	11111	Stakeholder 4: Maybe too colorful which is annoying. But it could work for students who like bright colors. Stakeholder 5: I think it's very bright and attractive. I think the colors are motivating and engaging. I really like it Stakeholder 2: Easy to read and see, engaging and doesn't take too much space!
3D color icons 		Stakeholder 2: Too distracting and too colorful. And if image shades and

		color are annoying. I don't like how they look. Stakeholder8: This looks a bit weird to me.
Real image 	1	Stakeholder 6: Too big and take too much spacing, take away what ur trying to learn. Stakeholder4: I like how it looks but if it's on a small scale then it would be quite ugly.

Design Idea Results (what did you end up selecting?)	
Design idea	Screenshot of selected option
Color Palette	
Web layout	 "Note: this image was selected only as a ref for information density and stat display, not as a literal layout to copy. The actual structure was developed and tested separately through hand drawn low-fidelity wireframes in Task 5. The final layout was simplified after low-fidelity testing showed the density

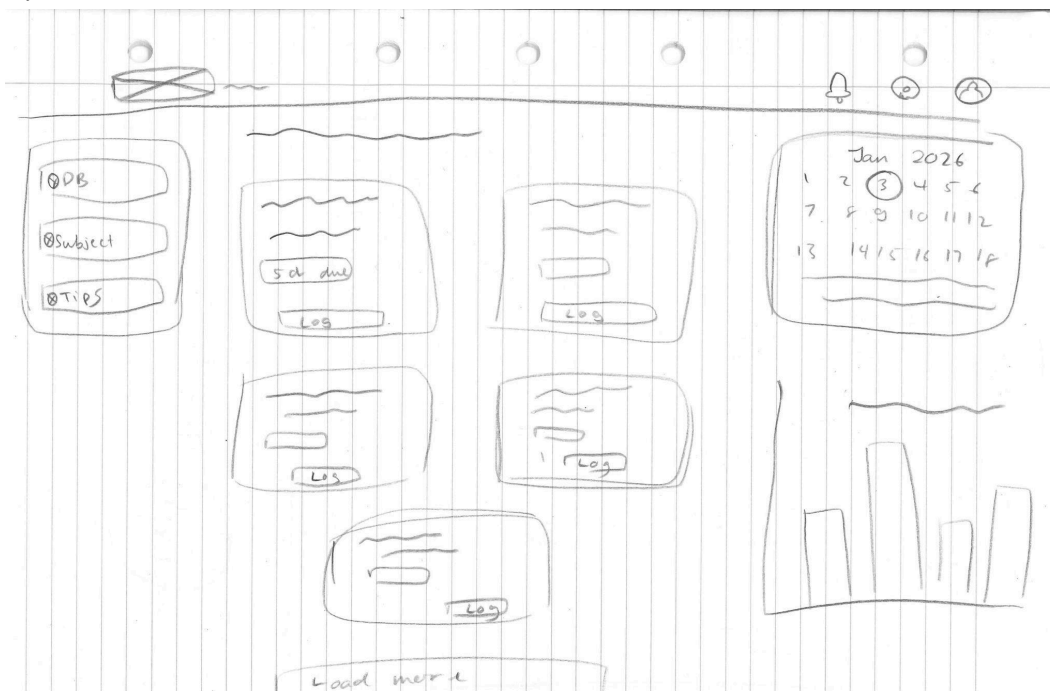
	<i>dashboard was overwhelming for users."</i>
Fonts	Regular 400 Daily priority list:
Images/Icons	

Task 5: Low fidelity

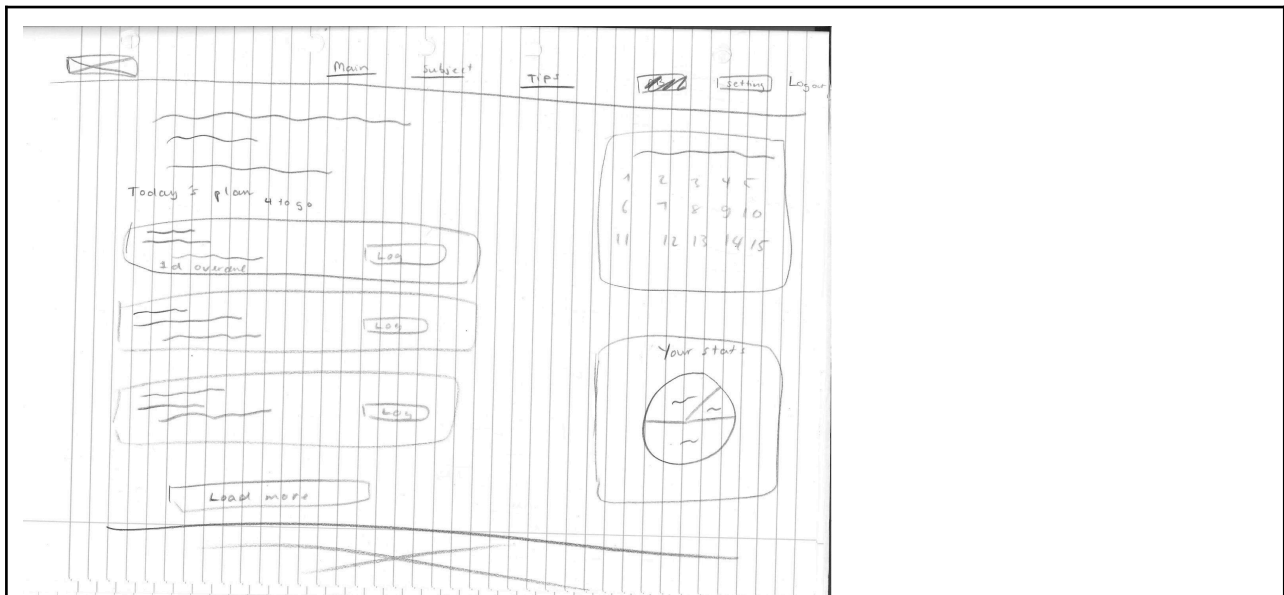
Screenshot of options:

(should be at least 3)

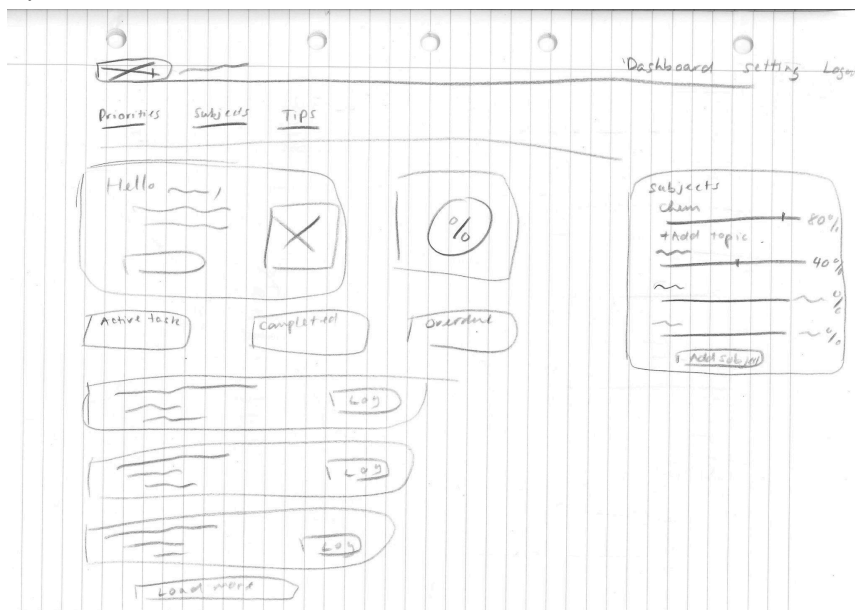
Option 1:



Option 2:



Option 3:



Stakeholder	Preference	Explanation
Stakeholder 1:	2	1. Navigation: For Design 1, I liked the overall side navigation concept but felt the empty spaces sitting below, so it looked unbalanced and slightly out of order and it made the page feel unfinished.

		Design 2's top navigation bar was much more preferred because it felt more familiar and functional and it's what most websites use, so it's easier to read and feels intuitive straight away. Design 3's navigation was actually my favourite for placement and structure, so ideally I'd want Design 3's nav style. 2. Settings: In Design 1, I really liked the header icons because they're clear and helps you understand the page quickly. For the settings, I strongly prefers the icon based approach from Design 1 rather than written text, because icons are faster to recognise. Design 2 &3 using text for settings feels slower to navigate and adds more distraction. 3. Layout : Design 1's layout of five subject boxes with a "load more" option work really well for me because it doesn't overwhelm you and gives just enough at once. And the bar graph from the homepage gives too much anxiety and distraction, it's not easing like percentage or pie graph. Design 2's spacing felt more logical though, and the overall order made more sense to follow. This is my most preferred one. Design 3 felt too crowded on the homepage because it had active tasks, completed tasks, and extra sections all in one place. I actually think that stuff belongs on the subject detail page/setting, not the homepage. Subject Progress / Hello Section: In Design
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		3, I liked the subject progress section being visible on the homepage, that is useful. The hello/account section in Design 3 is a friendly touch, but when I'm trying to focus on studying it becomes a bit of a distraction. Overall, I'd combine -> Design 3's navigation structure, Design 1's icon-based settings, Design 2's spacing and layout.
Stakeholder 2:	3	Navigation: I preferred the top navigation bar from Design 2&3 over the side dashboard in Design 1. The top bar felt cleaner and is easier to navigate. Design 2 had the navigation spreads across the top section which make me feels unstructured/distracted. I think if it's spread out too much it becomes harder to know where to look. So I find design 3 the best. Icons: I really liked the icons in Design 1, they make the interface easier to understand quickly. More icons and less plain text is better. So I would choose design 1. Plain text/ surrounded by buttons is not preferred to me. Layout: Design 1's box layout for showing subjects and tasks was clean and I liked the overall dashboard feel on the side. But the info inside the boxes felt a bit crowded and slightly overwhelming. Design 2 felt more organised overall. But design 3 combined design 2 with much more detail and features which I liked the best. Calendar: I think the calendar is necessary, it's useful to see when things are due and helps with planning. Design 1&2 had this included which I thought was good. Subject Progress / Hello Section: In

		Design 3 I liked the idea of the hello/account section, a lot of apps have something like that and it makes it feel more welcoming and personalised. However, the subject progress section in Design 3 was difficult to understand and I probably wouldn't use it.
Stakeholder 3:	1	Navigation: Design 3's navigation wasn't bad but felt a little unusual in placement, maybe just needs slight adjustments. Design 2 had the top nav which is nice, but if you include settings and log out in it as well it might become too much and feel cluttered. Design 1's nav was easy to use and the side placement means you can access it from anywhere on the page, which is practical. Overall preferred side navigation because of accessibility and convenience. Icons: I preferred more icons over text labels in all designs, icons feel less cluttered. I think surrounding navigation items with buttons to make them neater is also a great idea. Liked the icon approach in Design 1 particularly. Layout: I liked Design 1's layout because it doesn't feel too overwhelming and has enough information without going overboard. I think rounded/curved edges would make it look softer and more approachable. Design 2 also had potential to be good if you could switch between a fuller layout and a minimal one depending on what the user wants , as the idea of a flexible layout is appealing. I suggested that the calendar could be optional since some users might find it overwhelming. Subject Progress / Hello: In Design 3,

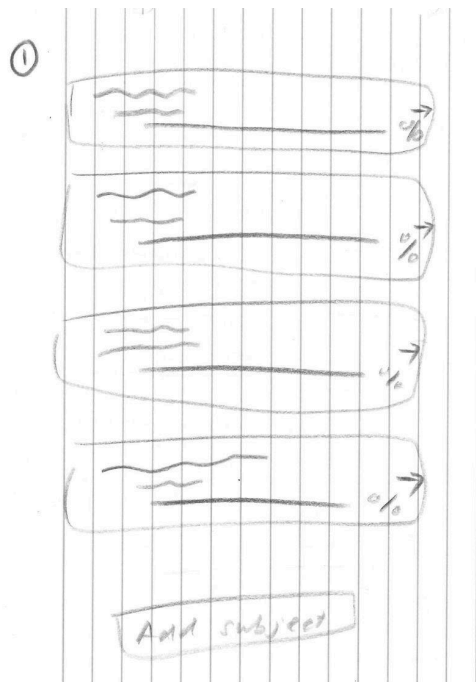
		liked the subject progress section, it's nice to see how much you've done. I also liked the hello/account section because it feels fun. Though this kind of detail makes the tool feel more engaging and student-facing.
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Item being tested: Subject detail intro page (the page that appears when a student clicks into a subject, showing a list of topics with navigation options)

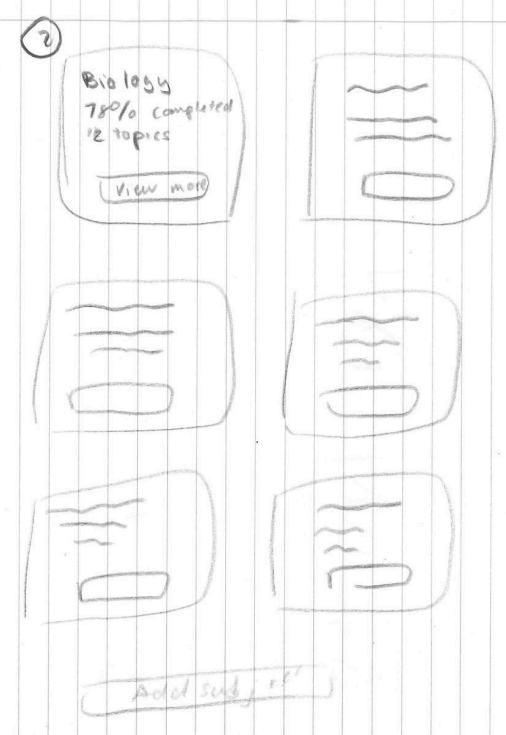
Screenshot of options:

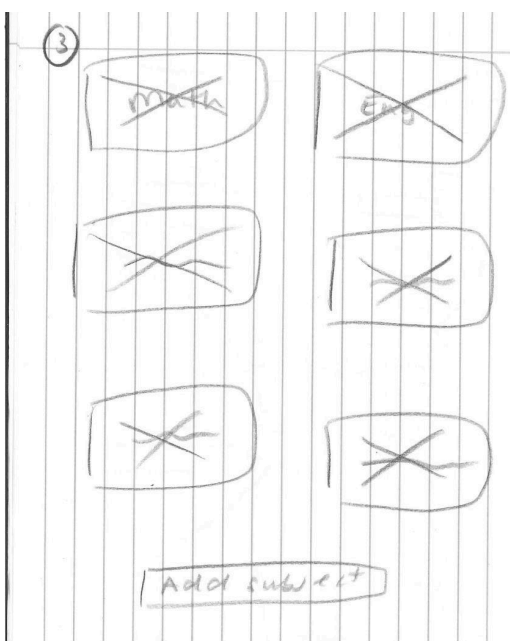
(should be at least 3)

Option 1:



Option 2:



		
Stakeholder	Preference	Explanation
Stakeholder 1:	1	Design 1: Really liked this design for the subject intro page. It felt clean, responsive, and easy to read without being distracting. The simplicity worked well here where you're going into a subject so you don't need to be overwhelmed with information. I would have liked slightly more detail shown per subject, but the overall layout was appropriate. Design 2: Also thought this was strong and gave clear information. The boxes in Design 2 didn't show enough detail though, which was a weakness. However, I liked the "view more" option in Design 2 over arrow navigation, as arrows in design 1 can easily be missed or overlooked by users, while "view more" is explicit and clearer. Because the homepage already uses a bar format, using the box layout here creates a nice distinction between pages. Design 3: The images in Design 3 felt too distracting for a study-focused page. The dark colour scheme also might not work

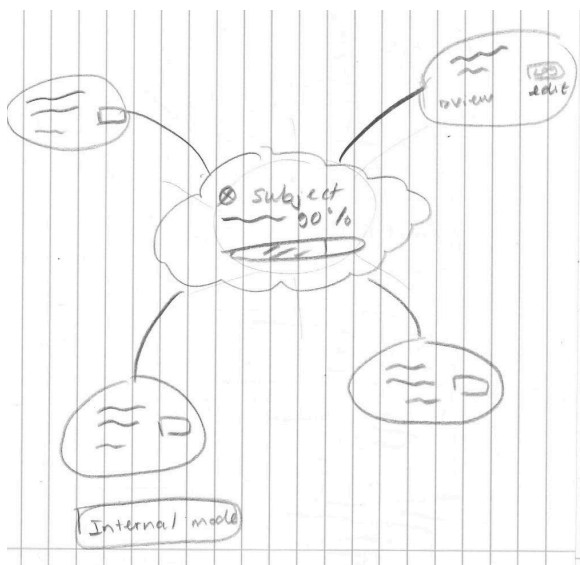
		for all users, like those with visual accessibility needs as it could reduce readability. Didn't feel it was appropriate for a study tool that needs to prioritise focus and clarity.
Stakeholder 2:	1	Design 1: I noticed that the shapes used in Design 1 for the subject detail intro were very similar to the homepage, which made it feel repetitive. Preferred the longer bar format on this page because it's easier to understand and scan. The arrow navigation was fine and felt intuitive. Design 2: Had too many boxes which felt overwhelming, the comment was that looking at that many boxes makes you not want to start studying. However, the "view more" button was preferred over arrows because it's clearer and easier to understand. Design 3: I strongly disliked this option. Said it looked like a food website rather than a study tool, and that images have no place in a subject detail view, they're distracting and pull attention away from the studying purpose.
Stakeholder 3:	1	Design 1: Thought this was good overall. She liked that it didn't overwhelm you with much information. She liked the bar format for this page and thought the arrow was fine. Design 2: Said the preference depends somewhat on which homepage design you go with as they should be consistent. Slightly preferred arrows over "view more" here because "view more" takes up too much visual space on the page. Also preferred the bar layout for this page over boxes.

		Design 3: Didn't think it was a bad concept, thought it could work for people who want a minimal-information view. But overall I felt it wasn't informative enough for a subject detail page, where you'd expect to see more content about what's inside.
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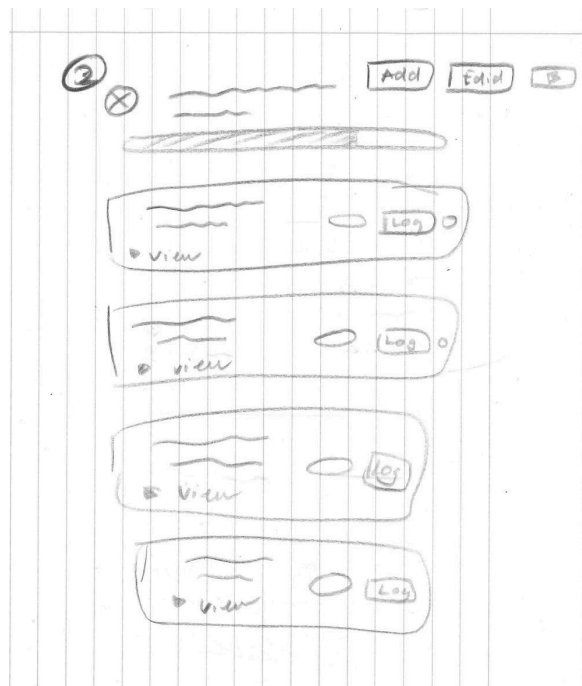
Item being tested: Subject detail topic page (the full page showing all topics within a subject, including priority ranking, review history, and progress tracking)

Screenshot of options:
(should be at least 3)

Option 1:



Option 2:



Option 3:



Note on Stakeholder Evaluation: A total of eight stakeholders (Stakeholders 1–8) participated in broad testing, where all eight voted on initial aesthetic options, while a focused subset of three (Stakeholders 1–3) conducted in-depth wireframe and usability testing.

Stakeholder	Preference	Explanation
Stakeholder 1:	3	Design 1 (Word Cloud): Strongly disliked the word cloud. I found it very hard to understand what was going on, it wasn't clear what the cloud was communicating or how to interact with it. The crowdedness of the word cloud pushed out more important information and made the page feel chaotic. Also pointed out that the internal mode button needed to be visible at the top, because if there are lots of topics and the button is lower down, you'd have to scroll back up to find it which is inconvenient. If there are many topics added, the word cloud could become completely unreadable and overwhelming. Overall said they didn't like Design 1 at all

		for this page. Design 2: Liked this design more. Key feedback was about consistency, if you're going to use shapes to represent topics, keep the shape consistent throughout (e.g. all bars, or all squares and not a mix). Changing shapes mid page is distracting and breaks the visual flow. I liked the progress bar concept. Design 3: Liked this the most because it separated things clearly into distinct sections like priority topics and completed topics. The graph included was great as it shows sort of visual progress. But would have preferred to see even more detail and suggested that graph filters could sit inside a settings area rather than cluttering the page.
Stakeholder 2:	3	Design 1 (Word Cloud): I disliked the word cloud. I think the concept is interesting but it's confusing and distracting in practise, it looks more like a brainstorming exercise than a structured study tool. It's not immediately clear what the word cloud is telling you or how to use it. Also I agreed that the internal mode button should be placed lower on the page so it doesn't interrupt the flow of content at the top. Design 2: For me, it looked too similar to the homepage. I liked the progress bar format and the bar based layout in general. The similarity to the homepage was the main downside. Design 3: This was my favourite. I find it's structured and clear where you can see everything you need in one view. The preference was for it to be mainly text-based rather than heavily visual with shapes and bars, because too many visual elements start to feel overwhelming. I

		liked that it felt different and more comprehensive than the other designs.
Stakeholder 3:	3	Design 1 (Word Cloud): She acknowledged it was a fun and interesting concept but said in practice it was quite distracting and hard to actually use. Agreed that the internal mode button worked better at the bottom of the page rather than the top. Design 2: Liked the overall progress tracking visible in Design 2. Noted that it was a bit similar to the other pages and suggested having some variety in shapes rather than everything being bars and boxes. Though having a breakdown of each individual topic was a good feature and should be kept. Design 3: This was her fav. She said having a full-page layout showing everything you need to know is very useful and well-organised. The comment was that if this design is chosen, it should start with content-heavy text at the top rather than leading with shapes and visual elements, because text communicates more directly what needs to be done. Liked that it separates from the boxes-everywhere pattern in the other designs.

Based on the feedback you collected, what aspects of your design worked or didn't work for your end users? Please summarise your feedback below:

Findings from feedback

Looking across all 3 pages and all 3 testers, there are some similar patterns that I noticed. For the homepage, the feedback split quite evenly across all three designs which tells me no design got everything right. What I can pull through though is that top nav (design 2 & 3) is preferred over side nav by 2 out of 3 testers. Icons are strongly preferred over text

labels across the testers (design 1) and calendar features are generally well received (design 1 & 2). Even though calendar features are generally well-received by users, as I reflect and consider all the feedback, I find that including one created unnecessary visual clutter and overloaded the dashboard. Stakeholder 8 noted that the calendar layout "does not strongly differentiate between important and less important revision tasks," which duplicated effort since the priority ranking list already handles that much better. Removing the calendar eliminated redundant information and prevented students from feeling overwhelmed. The subject progress section is wanted on the homepage (2 out of 3), while one of them finds it unnecessary and wouldn't really use it. The hello/account section has divided opinions where some found it welcoming and others found it distracting from the studying purpose. The main issue with design 3 was that the homepage was too crowded with features that belong to the subject detail/setting page.

For the subject detail like the intro page, all three testers preferred design 1 independently, which is a very clear result. My consistent finding was that this page needs to be clean, focused and not too overwhelming as you're transitioning into a subject, so the design needs to guide rather than distract. Images (design 3) were all rejected as inappropriate for a study tool. Design 2's boxes were seen as too repetitive and not informative enough.

For the subject detail page, all 3 testers prefer design 3, which is a clear winner. The word cloud in design 1 was universally disliked, as testers claim it looked unclear, distracting and more like brainstorming than study interface. Design 2 was seen as too similar to homepage and lacked variety. and the design 3 structured full page layout with separate sections was most preferred, and testers suggested it should lean more on text than shapes.

What am I continuing to implement based on feedback?

So based on the feedback, I'm keeping and refining several things. The top nav bar is going into the final design based on the majority preference, but centered below the top nav (that includes the logo & setting to prevent overcrowring). As it is more familiar and accessible. I'll keep the icon based settings rather than written labels since all testers prefer icons and it will add more visual engagement to the interface. The five subject layout as bars in design 2 is being carried forwards as it prevents overwhelming the users (since texts are quite separate in lines, with minimal overcrowding text). And the bars are curved/rounded edges because multiple testers said they made the interface feel softer and less static.

For the subject detail intro page, I'm keeping the bar/list layout from design 1 because

every tester preferred its clean, clear and readable layout. However, I would implement a view with more buttons instead of arrows as students may overlook the arrows and don't know where to navigate. So "view more" would give clearer instruction, which is important as my outcome aims to support every student with different technical/cognitive ability across nz.

For the subject detail page, I'm moving forward with design 3's structured section based layout because all 3 testers choose it. The separation of priority topics, reviewed topics and progress tracking into clearly labeled sections make most sense for how students would actually use the page. I positioned the Internal Mode button, which allows students to temporarily pause scheduled reviews for a subject during internal assessment periods, at a fixed position at the bottom of the screen so it remains easily accessible without requiring any scrolling. While Stakeholder 1 suggested placing it at the top, I followed the majority feedback from Stakeholders 2 and 3, who preferred the bottom layout. This decision also prevents visual overload at the top of the page, where the primary add, edit, and delete buttons are already located. So the Internal Mode feature stays prominent without cluttering the main nav controls.

Overall the feedback process from these three testers shaped the final design significantly more than I expected it. Before testing I assumed the dashboard-style layout from the colour palette research would be the obvious choice but Stakeholders 2 and 3 both independently flagged that too many boxes in one place made them not want to start studying, which is the exact opposite of what this tool is supposed to do. That feedback directly changed the final homepage from a grid to a limited list with "load more" option which is a big usability shift. The visual choices like colours and fonts changed the structure of how information is organised across the whole site. So the final design is genuinely different from what I would have produced without testing, which I think shows that the feedback process was actually useful to me.

What do I need to improve based on feedback?

There are some clear improvements coming out of this.

1. The word cloud is being completely removed, it was disliked by all 3 testers and didn't communicate anything useful. I'll replace it with a structure list layout from design 3.
2. The active tasks and complete sections are very nice but may need to move off the hp and onto the subject detail page where they belong, as having them on hp made it feel more clutter.
3. I need to simplify the subject detail page visually by leading with text-based content before shapes or graphs, since testers pointed out that too many visual

elements in 1 place becomes overwhelming.

4. For shapes and visual elements on the detail page, I'll keep the shape consistent, mixing circles, bars, and squares on one page was concerned as distracting.

So these improvements respond to what the testers said and will make the final design more focused and usable across different student types.

Task 6: High Fidelity

Wireframes / mock-ups:

Priority landing page:

Review Navigator

Setting

Account

Priorities

16 March 2026

Good afternoon Alex

8 overdue, 0 due today

Biology

Photosynthesis

- Reviews #2 You reviewed it 10 days ago

3d overdue

Log review ✓

Physics

Mechanics

- Reviews #1 You reviewed it 5 days ago

2d overdue

Log review ✓

Mathematics

Algebra

- Reviews #5 You reviewed it 15 days ago

1 day overdue

Log review ✓

+5 more topics when you're ready

Subject Details

Your subjects

Biology

75%

+Add topics

Mathematics

60%

+Add topics

English

60%

+Add topics

Physics

60%

+Add topics

+Add subjects

Subject Intro Page:

Review Navigator

 Setting  Account

Priorities

Subject Details

**Physics**
You have reviewed 3 out of 5 topics
60% [View More](#)

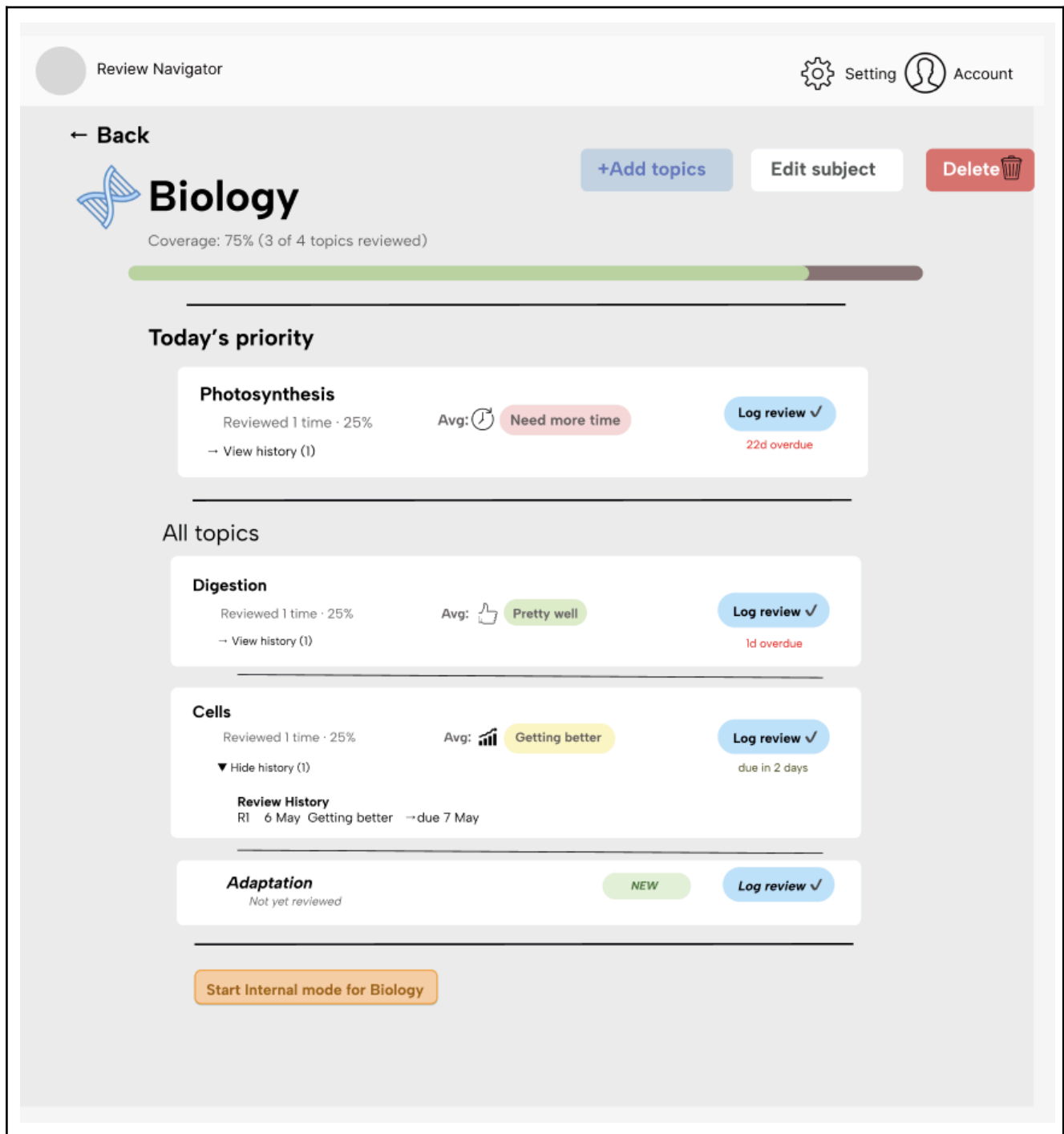
**English**
You have reviewed 3 out of 5 topics
60% [View More](#)

**Math**
You have reviewed 3 out of 5 topics
60% [View More](#)

**Biology**
You have reviewed 3 out of 5 topics
60% [View More](#)

[+Add subjects](#)

Subject Detail Page:



Why are the designs appropriate?

The purpose of Review Navigator is to help NCEA Level 2 and 3 students prepare for their end of year external assessments by automatically scheduling spaced repetition reviews and showing them exactly what topics to study each day. The problem it's solving is that students don't have a system for reinforcing content over time and they tend to learn something once, then not look at it again until exam time when it's mostly forgotten.

The high fidelity design I've selected reflects this purpose. The priorities page is the main view because that's the core daily interaction where you open the tool, you see what's overdue, you log a review, and you're done. The subject detail page gives a deeper view when a student wants to track a specific subject's progress. Both pages are designed to make the review process as fast and low-friction as possible which is the only way students will actually use it consistently. I used the initial decided palette as reference, however, as i design, i want to ensure a more minimal and balance color palette, so i shift toward a calmer blue with no cream, but it focuses on industrial standard/ world convection color like red, yellow, green that associate with meaning (e.g. red= delete/overdue) for student to easily understand.

At the start of the design process I identified several requirements the outcome had to meet.

1. The tool needs to show students what to study without them having to decide. The priorities page opens as the default view and immediately shows a ranked list of overdue topics, sorted by how long they've been waiting for review. Students don't need to navigate anywhere or make any choices as the algorithm has already done that. So this helps the outcome to be straightforward and reduce students' cognitive load as they navigate. The "+5 more topics when you're ready" link means they don't deal with every overdue topic at once, which would feel overwhelming. The outcome still gives students' autonomy to view more topics/log topics that they wanted, but showing 3 suggestions at a time can lower pressure load and make the overall interface less crowded/more aesthetic.

2. The tool needs to be usable for all students and used in under 30 seconds for a single review. Each priority card has a "Log review ✓" button on the right that's the only action needed. One tap records a review and the algorithm updates the next due date automatically. There's no length form to fill in, no rating required before logging, no navigation between screens for the basic interaction. And the homepage itself shows clear topics needed to revise today which students can instantly find the topics they wanted and log their review in a short amount of time.

3. The tool needs to be clean and uncluttered. The card-based layout with white backgrounds and plenty of white space keeps the page from feeling crowded -> students (end-users can easily navigate without getting confused/overloaded with info). The subject sidebar on the right gives a summary without adding more cards to the main column (so students don't have to scroll down more/or find it somewhere else that could lead to inefficiency/timewastage). The expandable "View history" in subject detail page keeps review data accessible without showing it by default (adding too much overwhelming info on one screen that can help end users to only see info that they actually need).

4. The tool needs to track progress and show it clearly. The 75% tasks completed stat on the homepage, the subject coverage percentages in the sidebar, and the "Coverage: 75%" on the subject detail page all give students different views of their progress. The progress bars make the percentage immediately visual rather than just a number. The bar itself can help students with difficulty of understanding words easily visualise the process and add more engaging elements to the design itse;f.

5.The tool needs to work across different devices. The web-based format means it opens in any browser on any device without downloading anything. The card layout adapts to different screen widths. This is taken into consideration of the end-users, as students use different devices to study.

Justified conventions:

Before designing, I looked at existing tools and applications to understand what conventions NCEA students already know and expect. I looked at Quizlet, Anki, Education Perfect, Duolingo, and study timet.

Top navigation bar: Every tool I investigated used either a top navigation bar or a side menu. Based on the low fidelity testing, 2 out of 3 testers said the top nav bar feels more familiar and easier to use than a side dashboard. This is consistent with what I saw in most webpages online where the top bar is the standard. I applied this in the final design by placing Priorities and Subject Details as tabs across the top, with the brand name on the far left and icons (settings and account) on the far right. This positioning is conventional across web applications and means students can find what they need without any learning curve.

Icon based settings and account: The icon for settings and the silhouette person icon for account appear in almost every major app students already use like Gmail, Spotify, Instagram, Google. Using these same icons means the design communicates instantly without labels. All 3 testers from the low fidelities testings all said they preferred the icon based settings over text labels, and this convention justified that preference as icons are faster to recognise because they don't require reading. However, the design also included text as well for clearer instructions and to avoid confusion with students who are unfamiliar with icons.

Colour-coded status indicators: I used red for overdue ("2d overdue"), green for solid confidence, and yellow for shaky confidence following the traffic light convention that students encounter in marking systems, road signs, and apps like Duolingo's streak warning system. I chose this because it communicates urgency and status without needing students to read the label. So a student scanning the page can immediately see which topics need attention by colour alone, though text labels are also included for accessibility .

Card-based layout: each priority item sits in its own card with a light border and rounded corners. This is a convention from task management tools like Todoist and Trello where individual items are visually separated so they feel complete. It also makes the interface easier to scan because each card is a self-contained unit of information. Stakeholder 3 from the low-fidelity testing said they liked the curved edges because they looked "softer and approachable", so the card convention also supports that aesthetic that end users looking for .

Progress bars for completion percentage. Every learning platform I looked at uses horizontal progress bars to show how far through a unit or subject a student is. Duolingo, Khan Academy, Education Perfect, and Quizlet all use them. Because students are already familiar with what a progress bar communicates, I didn't need to label or explain them as they can immediately understand that a full bar means complete and a partial bar means in progress. I applied this in both the subject cards on the priorities page sidebar and the coverage bar at the top of the subject

detail page.

Call-to-action button placement: The "Log review ✓" button is right aligned on every priority card. This can follow the convention in most apps where the primary action sits on the right side of a row, which you can see this in email clients like Gmail where archive/delete buttons sit on the right, and in to-do apps like Todoist. It's also how iOS swipe actions work on phones. Once students use the tool a couple of times, their hand goes straight to the right side of each card without thinking about it. Right aligning the action also keeps the left side free for the topic information, so the layout is scannable left to right where topic name on left and action on right which reduces the time it takes to process each card.

Task 7: Relevant Implications

Describe the implications that are relevant to your outcome. The following is a list of potential implications:

- social
- cultural
- legal
- ethical
- intellectual property
- privacy
- accessibility
- usability
- functionality
- aesthetics
- end-user considerations
- health and safety.

Implication 1

The implication of accessibility is relevant to my outcome because.... I need to make sure that any NCEA Level 2 or 3 student who needs this tool can actually use it, regardless of their circumstances. This matters because the tool is designed for a broad student population (all NCEA level 2 & 3) and not just students at high-decile schools with reliable devices, well cognitive ability and fast internet, but students across all school types and backgrounds. If the tool only works for students who have a personal laptop and a fast connection, it's not actually serving the problem it claims to solve.

Justify how selected design addresses relevant implications

The design addresses accessibility in several ways. First, the colour-coded system (red for overdue, green/amber for confidence levels) is always paired with text labels as well like the "2d overdue" text appears alongside the red colour, and the confidence tags say "shaky", "solid", "ok" in addition to their colour. This means the tool works for students with colour vision deficiency who might not distinguish red from green. Each piece of content is contained in separate cards that are spaced well apart from each other. This design improves accessibility for users who have poor eyesight or struggle with precise clicking, as the distinct sections prevent them from accidentally selecting the wrong button.

Secondly, all interactive buttons are large enough to be tapped accurately on a touchscreen like the "Log review ✓" buttons span the right side of each card which makes them easy to tap even on a small phone screen. The language used throughout the interface is plain and direct. I used languages like "Today's priority", "Log review", "Not yet reviewed", "2 days overdue" and none of these require prior knowledge of the tool or technical understanding to interpret. This makes the tool accessible to students with lower digital literacy or English as a second language, which is important given the diversity of the NCEA student population.

What I still need to improve for accessibility: the right sidebar on the priorities page needs to stack below the main content on mobile screens rather than sitting side by side, because on a small phone screen the two-column layout will make text too small to read. This is something to address in the next design iteration before coding.

Implication 2

The implication of usability is relevant to my outcome because....

Usability means students can use the tool easily and without frustration, and that it fits naturally into how they already study. This is probably the most important implication for my specific outcome because even a perfectly designed spacing algorithm is useless if students open the tool once and never come back. The design has to be genuinely easy to use and navigate.

Justify how selected design addresses relevant implications

Every major interaction in the design is kept as short and efficient as possible. Logging a

review is a single tap on the "Log review ✓" button with no lengthy text or multiple steps/buttons clicked first. Viewing your priorities is instant because it's the default landing page. It can help to reduce confusion as students don't have to navigate around just to view what they need to do that day (to improve efficiency).

Adding a topic happens from within the subject detail page with an "+Add topics" button. None of these require navigation between multiple screens to complete the core action. The two-tab navigation (Priorities and Subject Details) limits the total number of places a student can be. There's no nested menu system, no settings hidden under other settings, no hidden features that require discovery. Stakeholder 2 from the low fidelity testing said they found too many boxes "overwhelming and unmotivating" so the high fidelity design avoids this by using a clean list layout on the subject detail intro page rather than a grid of boxes. And a tester said "shapes should be consistent", so all 3 main pages consist of similar bar shapes with content contained inside. This can ensure that all pages are consistent with each other so users won't be confused/make mistakes due to unfamiliar/new layout introduced.

The "+5 more topics when you're ready" message on the priorities page is a specific usability decision as it prevents the page from feeling endless and gives students a sense that today's task is achievable. If all overdue topics were shown at once, a student with nine overdue topics might feel too behind to even start., which adds more cognitive loads. By showing three and indicating more are available, the tool encourages starting without causing overwhelm.

Stakeholder 1 from the testing feedback said the bar graph on the homepage "caused too much anxiety or distraction." I've addressed this by using the 75% tasks completed stat as the main progress indicator rather than a detailed breakdown graph as a single number is calmer and easier to interpret quickly.

Implication 3

The implication of aesthetics is relevant to my outcome because....

Aesthetics matters for this tool because the way it looks affects whether students feel calm or anxious when they open it, and whether they want to keep using it. Stakeholder 1 during low fidelity testing said the bar graph on the homepage caused "too much anxiety or distraction" and asked for it to be replaced with something calmer. Stakeholder 2 said they wanted something that felt "structured and clear" but not overwhelming. These responses show me that visual design directly affects how students feel about studying, which affects whether they'll use the tool at all.

Justify how selected design addresses relevant implications

The high fidelity design uses a white background with soft card shadows and rounded corners which creates a calm, clean feel. The colour palette is mostly neutral (like blue-toned palette (quite similar to option 2 in Task 4) with specific colours (red, orange, green) used only for status communication red for overdue, amber for shaky confidence, green for solid confidence/progress which is not dominant. This means colour is meaningful rather than decorative, which reduces visual noise.

The overall layout is organised with clear cards and content all wrapped in each card, to ensure that students distinguish what section/information they are reading. This helps the site to feel more organised and easier to navigate, while maintaining its visual uncluttered for end-users. The typography and font sizes use a clear hierarchy between subject names, topic names, and metadata text so students can scan a card and find what they need instantly.

And I removed the calendar from the final design because it duplicated information the priority ranking already provides, Stakeholder 8 noted the calendar layout 'does not strongly differentiate between important and less important revision tasks', which is the exact function the priorities list performs better. And I find it adds much more clutter to the dashboard homepage without bringing actual benefits, as it can overload users with lots of info and make them overwhelmed. This can ensure that my dashboard design is minimal and easy to navigate that supports aesthetics.

Implication 4

The implication of cultural and social is relevant to my outcome because....

Social and cultural acceptability means making sure the design doesn't unintentionally exclude, or misrepresent any group within the target user population. For my outcome, this matters because the tool is designed for NCEA students across New Zealand as a whole and not just students at high-decile urban schools, but students from Māori, Pasifika, immigrant, and rural communities who make up a significant proportion of the NCEA student population, particularly at Levels 2 and 3. So my outcome needs to make sure the content and the outcome itself has to be socially and culturally appropriate.

Justify how selected design addresses relevant implications

My first consideration is language. All text throughout the interface uses plain, direct English that avoids idioms, educational jargon, or metaphors that assume a particular

cultural background. Phrases like "Today's priority", "Log review", "2 days overdue", and "Not yet reviewed" communicate their meaning directly without requiring familiarity with any specific cultural reference. This is a deliberate choice because I considered using more casual or colloquial language such as "stuff to revise today" or "you nailed it", but decided against it because informal language that feels natural to some students can feel excluding or even patronising to others, particularly students for whom English is a second or additional language. If I had used casual, culturally specific phrasing, students from Maori, Pasifika, or recent immigrant communities may have felt the tool was not designed for them, which would reduce their engagement with a tool that is specifically intended to support all NCEA students equally across New Zealand. Using neutral, functional language is safer and more universally accessible.

The second consideration is representation. The design avoids using any photographic images of students, which removes the risk of representation issues completely. Photographs of people can unintentionally suggest the tool is designed for a specific group, and certain images or symbols can carry different meanings across cultures. Instead, I included a calculator icon for mathematics and the molecular icon for biology, which were collected from a universal educational icon set. These are universal, abstract symbols that are used across many existing educational platforms and do not carry cultural-specific meaning. The greeting on the homepage uses only the students first name, which they enter themselves during setup. The system does not require an English-language name and does not reject non-English characters, which means the tool works equally for students from any cultural background without requiring them to adapt to fit the design.

Implication 5

The implication of end-user consideration is relevant to my outcome because....

The end users of this tool are NCEA level 2&3 students across nz secondary school. This is a big group of students with different study habits, device used? preference?, levels of organisation and amounts of prior knowledge of how memory works. So the design has to work for students who are highly motivated and want progress data and also for students who are disorganised and need the tool to make decisions for them.

Justify how selected design addresses relevant implications

The design addresses end-user considerations by recognising that not all NCEA students

study in the same way or have the same needs. The tool is designed to work for two different types of students at the same time without forcing either group to interact with features they do not need.

The first type is the student who wants a quick daily check-in. They want to open the tool, see what to do, log a review, and close it. For this student, the priorities page is the default landing view and immediately shows a ranked list of overdue topics with a single-tap "Log review" button on each card. There is no navigation required, no decision to make, and no additional steps before completing the core action. This directly addresses the end-user requirement I stated in Task 1, which was that the tool must reduce decision making by prioritising what to study. If this student had to decide what to review themselves, the tool would be replicating the exact problem it is supposed to solve.

The second type is the student who wants to go deeper and check their progress across subjects. For this student, the subject detail page shows coverage percentages, individual topic review history, confidence averages, and full topic management options. This information is kept on a separate page so it does not clutter the priorities view for students who do not need it. One issue I had to think about was whether showing overdue topics and confidence levels would actually put students off using the website. Earlier research suggested that students can disengage when feedback makes them feel like they are failing. However, if I removed all indicators of urgency, students would not know what to revise first (which one is most urgent), which doesn't go with the purpose of the tool. Because of this, I tried to balance both needs. So instead of using judgemental language, the website uses terms like "ready for review" and confidence labels such as "shaky", "okay", and "solid". The colours are there to help students prioritise tasks quickly, but they are always paired with text so the focus remains on what to do next rather than on what the student has done wrong.

The low fidelity testing showed that both types of student genuinely exist among the target users. Stakeholder 3 said "it's nice to see how much you've done" and valued the subject progress details, while Stakeholder 1 said the homepage felt cluttered when it tried to show everything at once and suggested that detailed stats belonged on a different page. These two responses directly conflicted with each other, which showed me that I could not design one layout that would satisfy both users at the same time. The final design resolves this by compressing the progress overview into a sidebar summary on the priorities page and moving all detailed data to the subject detail page. This means the student who just wants to log a review is not overwhelmed, and the student who wants to track their progress can find that information without it being pushed on them by default.

Implication 6

The implication of privacy is relevant to my outcome because....

The implication of privacy is relevant to my outcome because during my research I found that Education Perfect makes student accuracy and engagement data being seen to teachers and, in some cases, to the class through leaderboards. This conflicted directly with what students said they needed, since seeing your own struggles made visible to others can discourage students who are already behind from engaging honestly with the tool.

Justify how selected design addresses relevant implications

Review Navigator avoids this by keeping all review history, confidence ratings, and overdue status visible only to the individual student logged into their own account. There is no leaderboard, no class visible accuracy data, and no teacher observing the learning in the design. This was a decision I made from Education Perfect's model, because a tool is meant to support consistent honest engagement, and needs students to feel safe logging an honest 'shaky' rating without it being seen by anyone else.

Task 8: Present Your Design

Please present your final design here.

Present your final design here and explain why you selected it as your final design by justifying your decisions with some of your low-fidelity work.

The final design of the review navigator consists of 2 primary pages: the priorities dashboard and subject detail page (the subject intro page is linked to the subject detail page).

Why is this design the right solution for these specific end users?

The end users I defined in Task 1 are ncea level 2 & 3 students who are willing to study but struggle with knowing what to review and maintaining motivation across a long academic year. These students have 2 modes: quickly daily check ins (e.g. open, log review and close) and deeper progress views (e.g. check subject coverage, manage topics...). The final design addresses both modes without forcing either group to interact with features they don't need.

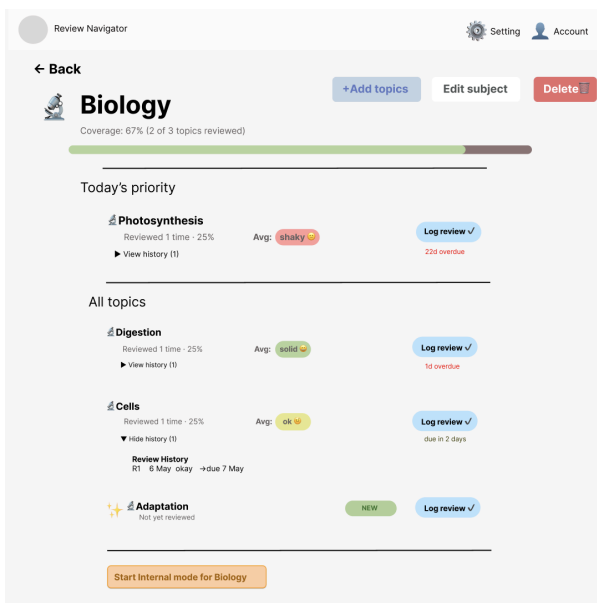
The priorities page addresses the quick-check user. When a student opens the tool, they immediately see a ranked list of overdue topics, a single-tap "Log review ✓" button, and a daily completion percentage. And there's no decision required as the algorithm has already prioritised what to do. This addresses the end-user requirement I identified in Task 1: "The tool must reduce decision-making by prioritising what to study." If the tool required students to decide what to study themselves, it would be replicating the problem it's supposed to solve, and they'd stop using it. The low fidelity feedback confirmed this need as Stakeholder 1 said the homepage felt cluttered when it tried to show everything at once and that "stuff belongs on the subject detail page, not the homepage." That feedback directly shaped the decision to keep the priorities page simple and push detailed progress information to its own page.

The subject detail page serves the deeper-review user. It shows coverage percentage, review history per topic (expandable so it doesn't crowd the default view), confidence averages, and individual topic management (add, edit, delete). Stakeholder 3 from the low fidelity testing said "it's nice to see how much you've done" and liked the subject progress details so I kept all of that information accessible, but on a separate page where it doesn't interfere with the daily interaction. Stakeholder 1 said the opposite, where they found detailed stats distracting from the homepage. So by separating the two views, the final design actually balances both students at the same time, where the one who wants data and the one who just wants to be told what to do.

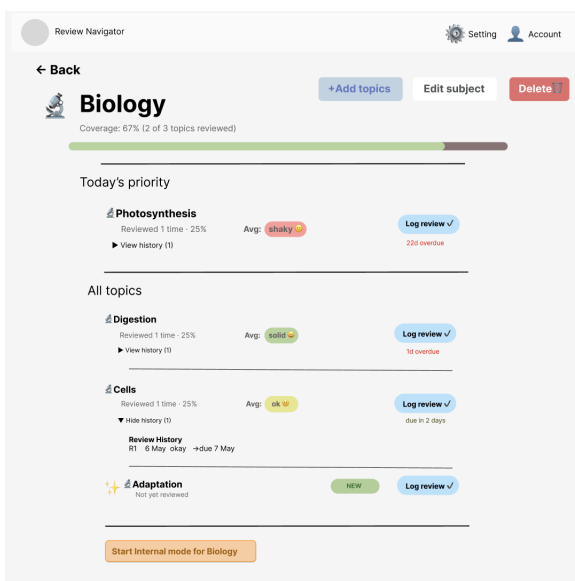
How low fidelity changes/shape the final design

The final design is different from what I would have designed without feedback.

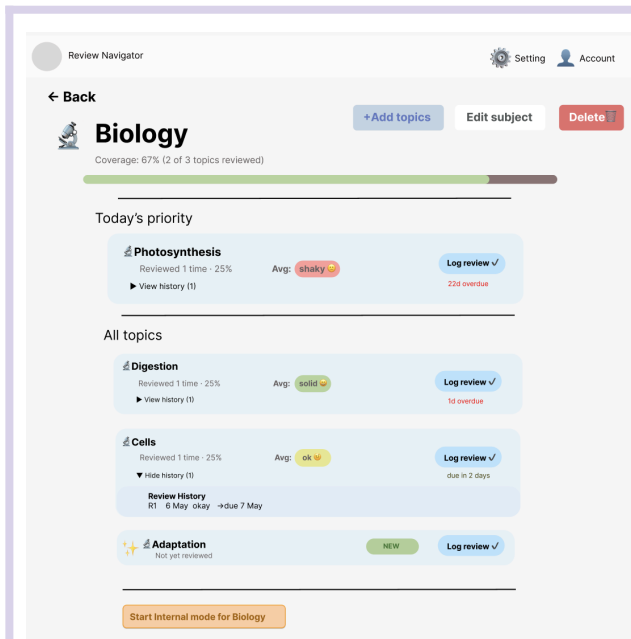
One piece of iteration that happened beyond the three original wireframe options was in the subject detail topic page layout. In the initial pencil sketch wireframes, the layout used only dividing lines between topics because the design looked clean and uncluttered at that stage. However, once I started placing actual content into the design, including topic name, review count, percentage, self-rating, overdue status, and review history, the lines-only layout did not create enough visual separation between items. The information started to blur together.



I made a second version where I added light lines around each topic while keeping the dividing lines.



This also did not work because having two different types of separator competing with each other made the page feel inconsistent rather than clearer. In the third version, which became the final high fidelity design, I used clean rounded card boxes with a light background for each topic, matching the card style from the priorities page.



When I showed this version to the same testers, two out of three changed their preference and said this was the clearest version, and the remaining tester acknowledged the readability had improved. I moved forward with the card layout even though it was opposite to what those testers had said they wanted at the pencil sketch stage. This shows that design iteration that goes beyond the initial testing round can produce better outcomes than simply implementing first-round feedback directly.

The word cloud from wireframe Option 1 of the subject detail page was completely removed after all three testers rejected it independently. Stakeholder 2 called it "very hard to understand", and "looks more like brainstorming than a study tool". And Stakeholder 3 said it was "distracting in practice." I replaced it with Design 3's section based layout (Today's priority / All topics / Coverage bar) which all three testers preferred. Without that feedback round I likely would have kept the word cloud because I thought it was visually interesting, but the users made it clear that interesting isn't the same as useful for a study tool that needs to be immediately understandable.

The top navigation bar came from testing. Before testing, I had designed Option 1 with a side dashboard because I thought it looked like a proper web app. But 2 out of 3 testers said the top nav in Options 2 and 3 felt "more familiar and functional" and one tester specifically said that the side nav "looked unbalanced and unfinished." Combined with my research finding that every major educational tool uses top nav on mobile first designs (Quizlet, Duolingo, StudyTime), the feedback made it clear that the top nav was the right convention for this end user.

The icon-based settings came from all three testers independently expressing the same preference for icons over text labels in the navigation area. Stakeholder 2 said "more icons

and less plain text is better" and Stakeholder 3 said "icons feel less cluttered." This was also consistent with my research as every major app that NCEA students already use daily (Gmail, Instagram, Spotify, Google) uses icon based navigation rather than text menus. Because students already know what a gear icon means and what a person's silhouette means, using those icons removes the need to read anything to navigate, which reduces friction in a tool that students are supposed to use quickly.

How the final design addresses the requirements in Task 1:

The final design addresses all 5 requirements i identified in Task 1:

1. **Requirement 1 was that the tool shows students what to study without them having to decide.** The priorities page does this as its main function where topics are ranked by how long they've been waiting for review, the most overdue appear first, and the only action on the page is "Log review." Students don't navigate to find this information, it's the default view they arrive at. The "+5 more topics when you're ready" link means even students with many overdue topics aren't overwhelmed on opening the app as they show a manageable starting point.
2. **Requirement 2 was that the tool be usable in under 30 seconds for a single review.** Logging a review is one tap with no forms/additional buttons clicked before logging. The homepage shows clearly labeled topic cards, each with a single action button on the right side. Stakeholder 5 from the Task 4 layout voting said they liked "seeing all my progress and tasks immediately because it makes studying feel more manageable and motivating," so the default view was designed with exactly this in mind.
3. **Requirement 3 was that the interface is clean and uncluttered.** The card-based layout with white backgrounds and white space between cards prevents a lot of element crowding. I specifically chose not to include a progress graph on the homepage after Stakeholder 1 said a bar graph would "cause too much anxiety or distraction" but instead I use a single percentage (75% tasks completed). I chose a single percentage over a progress chart because Stakeholders 2, 3 and 5 all said a page dense with statistics felt overwhelming at the low-fidelity stage, and because a single number can be read in under a second, which supports the under-30-second requirement. Detailed progress data is kept on the subject detail page for the students who want it. The sidebar on the priorities page gives subject summaries in a column rather than adding more cards to the main feed, so students can see their subjects without scrolling.
4. **Requirement 4 was that the tool track progress visibly.** The 75% completion stat, the subject coverage percentages in the sidebar, the "Coverage: 75% (2 of 3 topics reviewed)" header on the subject detail page, and the individual progress bars across both pages give students four different views of the same progress data at different levels of detail. The colour coded progress bars (which the majority of

stakeholders voted to retain during layout testing) make the percentages immediately seen where students with lower literacy or numeric processing difficulties can read the bar without needing to interpret the number.

5. **Requirement 5 was that the tool work across different devices.** The web based format means it runs on any browser without downloading or installing anything, which matters for students who switch between home devices, school computers and phones. The card layout stacks vertically on narrower screens though the sidebar layout on the priorities page still needs to be made fully responsive for small phone screens before the tool goes into production.

Accessibility is addressed through free access, web-based format, colour + text label pairing (so colour vision-impaired students aren't relying on colour alone), plain language throughout, and large tap targets on the Log review buttons. Usability is addressed through the single-tap interaction, the two-tab navigation with no nesting, and the limited topic display with a "more" option. Aesthetics are addressed through the calm blue-toned palette chosen by stakeholders, the consistent card treatment, and the avoidance of visual elements that caused tester anxiety (bar graphs, word clouds, dark colour schemes). Cultures and social responsibility are addressed through plain direct english language and use of icons to avoid any culture mis-inpretation images. End-user consideration is addressed through compressing and simplified priorities layout while keeping the detailed information for different end-users preferences.

And I do note that this design isn't perfect. I've acknowledged in Task 7 the places where more work is needed, particularly making the priorities page stack properly on mobile screens before this goes into production. These are limitations I'm aware of and the core design decisions throughout this portfolio are collected and synthesized in the research, convention analysis, and user feedback documented across Tasks 2 to 6.